

TheMakerVoid ELS v1.0

Electronic Lead Screw for Metal Lathes — User Manual

ESP32-S3 • ILI9341 320×240 Touchscreen • Lichuan A6 Servo • 1440 p/rev Spindle Encoder

1. Introduction

The **TheMakerVoid ELS** (Electronic Lead Screw) is a controller that electronically couples your lathe's spindle to the carriage drive motor. Instead of swapping physical change gears, the controller reads the spindle rotation through a rotary encoder and commands a servo (or stepper) motor to move the carriage at a precisely synchronized ratio. This lets you perform **automatic feeding** and **thread cutting** — in both metric and imperial — and use the spindle as an **indexing / dividing head**, all selected from a touchscreen menu.

The unit is operated either by the **resistive touchscreen** or by **four physical buttons** (UP, DOWN, SELECT, BACK). Both input methods are always active and do exactly the same things.

Who this is for: machinists and hobbyists fitting an ELS to a manual lathe. Basic familiarity with lathe feed rates, thread pitch, and TPI is assumed.

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2. Safety

Read before use. An ELS commands carriage motion automatically whenever the spindle turns in Feed or Thread mode. Carelessness can crash the carriage into the chuck or your hands.

- Always know which **direction** the carriage will travel before engaging. Verify the CW/CCW direction setting (Section 11.1) for your machine.
- Keep the **BACK** control within reach. Pressing BACK from Feed or Thread mode immediately stops the motor and returns to the Main Menu.
- Confirm the configured **lead screw pitch** (Section 11.2) matches your lathe's actual lead screw. A wrong value produces a wrong feed/thread ratio.
- Make a trial pass clear of the workpiece the first time you use a new feed rate or thread pitch.
- The motor only moves while the spindle moves. Stopping the spindle stops the carriage.

3. Hardware Overview

Component	Specification
Microcontroller	ESP32-S3 DevKit C-1 (dual-core)
Display	ILI9341 TFT, 320×240 px, landscape, resistive touch (XPT2046)
Spindle encoder	Quadrature rotary encoder, 1440 pulses per revolution
Motor driver	Lichuan A6 servo drive, PUL/DIR (step & direction) interface
Physical input	4 buttons: UP, DOWN, SELECT, BACK
Persistent memory	EEPROM — stores direction, lead screw pitch, and dark mode

Internally the controller uses both ESP32 cores: one core drives the user interface and screen, while a second dedicated high-priority core does nothing but read the encoder and generate motor pulses in real time. This separation keeps motion accurate and jitter-free even while you navigate menus.

4. Controls

4.1 Physical Buttons

Button	General function
	Move selection up / previous item / increment a digit
	Move selection down / next item / decrement a digit
	Confirm / enter the highlighted option / advance
	Go back one level. From the Main Menu it opens the Settings overlay; from Feed/Thread mode it stops the motor and returns to the Main Menu.

4.2 Touchscreen

Every on-screen button, list item, and arrow can be tapped directly. The screen recognizes one touch at a time and ignores repeats for a fraction of a second to prevent accidental double-taps. Common on-screen icons:

Icon	Meaning
Menu (top-right)	Opens the Settings overlay (Configuration / Interface / Info).
Back (bottom-right)	Returns to the previous screen / cancels.
OK (bottom-right, left of Back)	Confirms the current selection or saves the setting.
Reload (Divider screen)	Resets the angle reading to zero.
 /  carets	Previous / next page in long lists (thread selection).
Stepper ▲/▼	Increment / decrement the selected digit (lead screw pitch).

5. Powering On

When power is applied, the controller shows a **splash screen** with the TheMakerVoid logo for about 5 seconds. You can skip it immediately by touching the screen or pressing any button. After the splash, the **Main Menu** appears.

On first power-up the controller initializes its saved settings to safe defaults (direction = CW, lead screw pitch = 3 mm, dark mode = on). On every subsequent boot it restores whatever you last saved.

6. Main Menu

The Main Menu is the home screen. It shows the three operating modes on the right and the **live spindle RPM** at the bottom-left, which updates continuously as the spindle turns (so you can use it as a tachometer at any time).

Option	Goes to
FEED	Feed-rate selection → automatic feeding (Section 7)
THREAD	Metric/Imperial choice → pitch selection → thread cutting (Section 8)
DIVIDER	Live spindle angle / indexing readout (Section 9)

Navigation: use **UP**/**DOWN** to highlight a mode and **SELECT** to enter it, or simply tap it. Press **BACK** (or tap the Menu icon) to open the Settings overlay.

7. Feed Mode (Automatic Feeding)

Feed Mode drives the carriage at a fixed distance per spindle revolution — the electronic equivalent of engaging the feed via change gears.

7.1 Selecting a feed rate

From the Main Menu choose **FEED**. A grid of available feed rates (in mm per revolution) appears. The five rates are:

0.07 • 0.10 • 0.13 • 0.18 • 0.25 mm/rev

Highlight a rate with / (or tap it), then press / tap **OK** to engage.

7.2 While feeding

The Feed Mode screen shows the active feed value and the **live RPM**. As soon as the spindle turns, the controller moves the carriage at the selected rate. Stopping the spindle stops the carriage.

To stop feeding, press or tap **Back**. This **immediately disables the motor** and returns you to the Main Menu.

Feed direction follows the configured motor direction (Section 11.1) combined with the spindle's rotation direction.

8. Thread Mode (Thread Cutting)

Thread Mode synchronizes the carriage to cut a thread of a chosen pitch. It supports both metric pitches and imperial (UTS) threads specified in TPI.

8.1 Choosing the thread standard

From the Main Menu choose **THREAD**. The **Thread Unit Switch** screen asks which standard you want:

Option	Meaning
METRIC	Pitch specified in millimetres (mm).
UTS	Imperial / Unified threads specified in threads per inch (TPI).

Toggle with / (or tap), then confirm with / **OK**.

8.2 Selecting the pitch

A paginated grid of pitches appears (six per page). Use the / **carets** at the bottom-left — or keep pressing /, which scrolls across pages automatically and wraps around —

to find your pitch. The header shows the units in use, (mm) or (tpi).

- **Metric:** 16 pitches from 0.20 mm to 4.00 mm.
- **Imperial (UTS):** 21 values from 8 TPI to 56 TPI.

(Full lists in Section 13.) Highlight the pitch and press **SELECT** / tap **OK** to engage.

8.3 While cutting

The Thread Mode screen shows the active pitch (in mm) or TPI value plus the live RPM, and the carriage tracks the spindle at the exact ratio for that thread. As with feeding, press **BACK** / tap **Back** to stop the motor and return to the Main Menu.

Important: the controller maintains the correct lead ratio, but thread cutting still requires correct technique — appropriate spindle speed, retracting the tool at the end of each pass, and re-engaging at the same point. The ELS keeps carriage motion locked to the spindle; it does not automate tool infeed or thread-stop.

9. Divider Mode (Indexing)

Divider Mode turns the spindle encoder into a precise **angular position readout**, useful for indexing work such as bolt-hole circles, flats, gears, or graduations.

- The screen shows the current spindle angle in **degrees (0–360°)**, updated in real time as you rotate the spindle by hand.
- Tap the **Reload** icon (or press **SELECT**) to **zero the angle** at the current position — set your reference, then rotate to the angle you need.
- Press **BACK** / tap **Back** to return to the Main Menu.

Because the encoder has 1440 pulses per revolution, the readout resolves to a quarter of a degree.

10. Settings Menu (Overlay)

From the Main Menu, press **BACK** or tap the **Menu** icon (top-right of most screens) to open the Settings overlay. It offers three items:

Item	Purpose
CONFIGURATION	Set motor direction and lead screw pitch (Section 11).
INTERFACE	Toggle dark mode (Section 12.1).
INFO	Firmware version and credits (Section 12.2).

Navigate with **UP**/**DOWN** + **SELECT**, or tap. **BACK** / **Back** returns to where you came from.

11. Configuration

These two settings define how the controller calculates every feed and thread ratio. They are stored permanently in EEPROM and survive power cycles. Set them up once for your machine.

11.1 Direction (CW / CCW)

Selects the motor rotation direction that corresponds to forward carriage travel on your lathe.

- Tap **CW** or **CCW** (or toggle with **UP**/**DOWN**).
- Confirm with **OK** / **SELECT** to **save**; the value is written to memory and applied to the motor immediately.
- **Back** cancels without saving.

If the carriage runs the wrong way when you engage Feed or Thread mode, return here and switch this setting.

11.2 Lead Screw Pitch

Tells the controller the pitch of your lathe's lead screw, in millimetres. This is the reference the ratio math is built on, so it must match your actual machine.

- The value is entered as **two digits** (tens and units), range **1–10 mm**.
- Tap a digit box to select it, then use the on-screen **▲/▼ stepper** arrows — or **UP**/**DOWN** — to change that digit (digits wrap 0...9).

- Press **OK** / **SELECT** to advance from the tens digit to the units digit; pressing it again on the units digit **saves** the value and returns to the Configuration menu.
- **Back** cancels.

12. Interface & Info

12.1 Interface — Dark Mode

The Interface screen has a single **DARK MODE** toggle. Tap the slider (or press **SELECT**) to switch between the dark theme and an inverted (light) theme; the change is shown instantly. The setting is saved when you leave the screen with **Back**.

12.2 Info

The Info screen lists firmware and hardware identification and project credits:

- TheMakerVoid ELS v1.0
- ESP32-S3 DevKit C-1 • ILI9341 TFT 320×240
- Lichuan A6 Servo • Encoder 1440 p/rev
- An open-source / public-domain project — free to use as you wish.
- Project & donation links to **www.themakervoid.com**.

13. Reference Tables

13.1 Feed rates (mm / revolution)

#	1	2	3	4	5
mm/rev	0.07	0.10	0.13	0.18	0.25

13.2 Metric thread pitches (mm) — 16 values

0.20	0.30	0.40	0.50	0.60	0.70	0.80	1.00
1.25	1.50	1.75	2.00	2.50	3.00	3.50	4.00

13.3 Imperial (UTS) threads — TPI — 21 values

8	9	9.5	10	11	12	14
16	18	19	20	22	24	28
32	36	38	40	44	48	56

13.4 Default settings (first power-up)

Setting	Default
Motor direction	CW
Lead screw pitch	3 mm
Dark mode	On

14. Troubleshooting

Symptom	Likely cause / remedy
Carriage moves the wrong way	Reverse the Direction setting (Section 11.1).
Feed/thread pitch is wrong by a constant factor	Check the Lead Screw Pitch value (Section 11.2) matches your machine's lead screw.
RPM reads 0 while spindle turns	Encoder wiring / coupling problem; confirm the encoder turns with the spindle.
Motor does not move in Feed/Thread mode	Make sure you confirmed the selection with OK /SELECT (the mode must be <i>engaged</i> , not just selected); verify motor driver power and PUL/DIR wiring.

Touch is offset or unresponsive	The touchscreen needs panel-specific calibration; have it recalibrated in firmware.
Settings reset after power cycle	You must press OK /SELECT to <i>save</i> a setting; leaving with Back discards Direction/Pitch changes.
Carriage keeps moving after you want to stop	Press BACK from Feed/Thread mode — this disables the motor. Stopping the spindle also stops motion.